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Engineering Portfolio

Mechanical design • Structures • Composites & manufacturing

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I design, analyse, build and test load-bearing structures. Most of that work has been on rocket hardware (composite airframes, machined metallic interfaces, a pressure vessel), because student rocketry's entire design is done by undergraduates: requirements, sizing, finite element analysis, tooling, manufacturing, testing and flight. Internships in aerostructures at Airbus and in advanced structures at Bombardier, and undergraduate research on polymer actuators for architected metamaterials, have extended the same interest in how structures and materials behave and how far they can be pushed.

This portfolio documents the engineering behind that work rather than repeating my CV: the requirements each design answered, the decisions and trade-offs, the analysis and how it was checked, and what testing and flight showed.

Project index

	Project and context	Methods	Key evidence	p.
1	**Project Luminor aerostructures** Hybrid rocket, McGill Rocket Team; lead of a ∼50-student subteam	Architecture, composite design, tooling and prepreg layup, FEA, CNC interfaces, integration	3rd place, Launch Canada (Advanced); structure recovered undamaged; 13.35 kg vs. 18.1 kg budget	2
2	**Classical laminate theory design tool** Individual project, used to size Luminor's composites	CLT (ABD), six failure criteria, shell and panel buckling, independent verification	24 benchmark quantities within 0.05% of an independent Python code; sized laminates flew without damage	7
3	**Structural redesign and optimization** Two flight components for a higher-apogee vehicle	Load-path redesign, topology optimization, design for 3-axis machining, mesh convergence	Recovery assembly 85% lighter; cage bending FoS 1.50 → 2.10 with 17.7% less mass	10
4	**Research: supercoiled polymer muscles** Architected Metamaterials Group; undergraduate RA	Fabrication, electro-thermal testing, failure analysis, integration design	12% stroke against a 2.84 N load (2.0 N required)	14
5	**Aircraft conceptual design** AERO 520 long-range business jet; team of four	Constraint sizing, performance, stability, structural layout	Field lengths 5,234 / 5,206 ft vs. 6,000 ft limit; 1.9% range shortfall traced to cruise L/D	17

Current work (2026–27): Design and Optimization Advisor, McGill Rocket Team; capstone project on the design, manufacture and ground testing of a one-piece composite fin can for the team's supersonic two-stage vehicle.

Software used in this portfolio: Siemens NX, SolidWorks, CATIA (CAD); Nastran, Ansys (FEA); MATLAB, Python, Excel; OpenRocket, RocketPy; XFOIL, AAA (Advanced Aircraft Analysis).

1 Project Luminor: Rocket Aerostructures

McGill Rocket Team • Aerostructures Subteam Lead, May 2025 – Aug 2026 • Subteam of ∼50 students

Vehicle. Single-stage hybrid rocket, significantly larger than the team's previous vehicle, built as forward, propulsion and aft assemblies.

My role. I co-led the aerostructures subteam from architecture through manufacturing, integration and flight, coordinating structural decisions with propulsion, avionics, recovery, payload and launch operations.

Outcome. Luminor placed 3rd in the Advanced category at Launch Canada, and the structure was recovered with no damage. Aerostructures mass was 13.35 kg against an 18.1 kg budget (26% under), and the full vehicle was repeatedly assembled in under one hour.

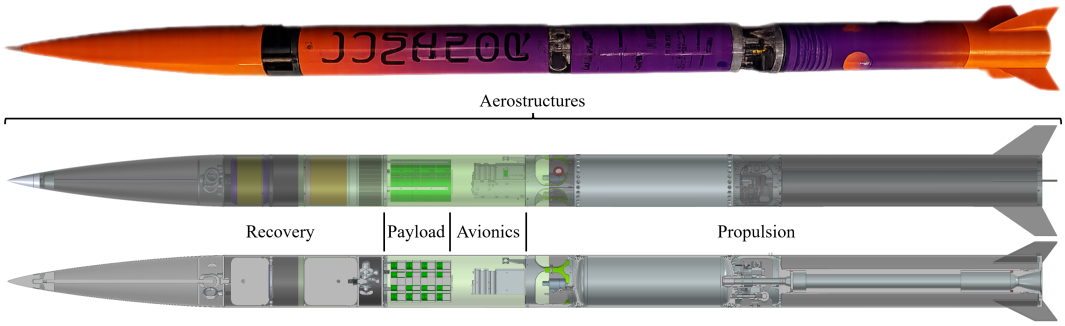


Figure 1: Project Luminor: final vehicle, CAD architecture and structural layout.

Problem and requirements

The previous competition showed that field assembly was a weak point, and Luminor's larger airframe raised both loads and mass. The structural subsystem therefore had to satisfy four coupled requirements:

1. Carry flight, propulsion and recovery loads with a minimum factor of safety of 1.5.
2. Stay within an 18.1 kg aerostructures mass budget.
3. Keep the vehicle's static stability margin within the lower and upper limits across the expected wind range.
4. Allow the vehicle to be manufactured, transported and assembled as independent modules.

Architecture and key design decisions

The external airframe is composite; internal structures and all primary joints are machined aluminium, with additively manufactured parts limited to non-primary structure. The table records the decisions that shaped the subsystem and the evidence behind each.



External airframe *Internal structures*

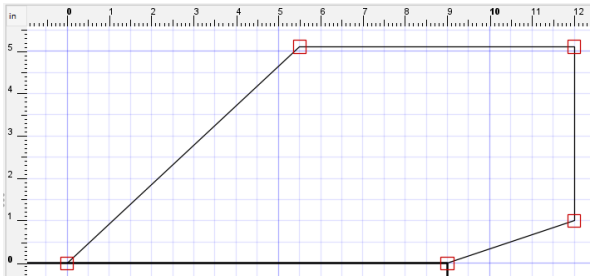
Figure 2: Aerostructures architecture: CFRP/GFRP airframe, honeycomb fins, RADAX couplers, payload and propulsion interfaces.

Decision	Rationale	Evidence
CFRP for most of the airframe; GFRP around avionics	Stiffness and mass efficiency where they govern; radio transparency where antennas sit	Laminates sized with the CLT tool
CFRP / aramid-honeycomb sandwich fins, swept clipped delta	High bending stiffness at low mass; planform iterated against static-stability and flutter constraints	Stability margin ≥ 1.5 calibers in winds up to $\sim$17 mph
Machined Al 6061-T6 RADAX couplers replacing the tube-and-leg joint	Higher joint stiffness, fewer structural parts, faster assembly	FEA: bending FoS 1.50, ejection FoS 2.82
Modular forward / propulsion / aft assemblies	Independent manufacture and transport; shorter assembly time	Assembly under 1 h in repeated rehearsals
Drogue deployment triggered before apogee	Lesson from the previous vehicle; gives the parachute time to inflate	Dynamic model $\rightarrow$ 30.6 m/s velocity trigger

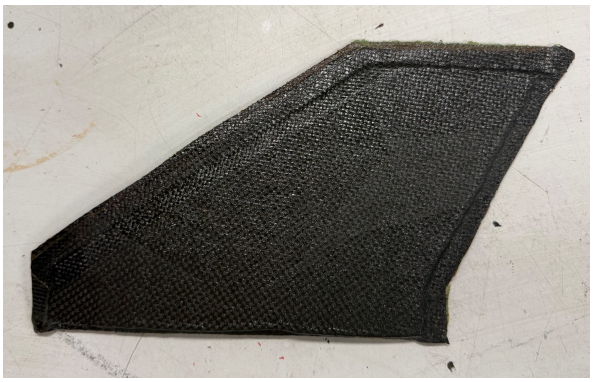
Composite design and manufacturing

The fins use three CFRP plies on each face of an aramid honeycomb core. Aerodynamic loads enter the airframe through a through-the-wall attachment, structural epoxy and continuous tip-to-tip reinforcement. Airframe tube laminates were sized with classical laminate theory.

The team produced its own tooling. For the nose cone I worked through the complete chain: CAD pattern $\rightarrow$ large-format PLA print $\rightarrow$ glass-fibre wet-layup female mould (sealed and release-treated) $\rightarrow$ CFRP prepreg layup $\rightarrow$ vacuum consolidation $\rightarrow$ oven cure. Airframe tubes were laid up on reusable aluminium mandrels and consolidated with shrink tape; flat laminates were vacuum-bagged and machined into internal parts, extending composite construction beyond the primary airframe.



Final fin geometry



Cured honeycomb-sandwich test fin

Figure 3: Composite fin design and manufacture.

Structural analysis

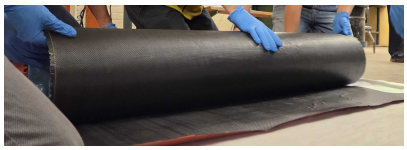
Each primary component was sized by hand calculation and checked by FEA under its governing cases: axial thrust and acceleration, flight bending, aerodynamic fin loads, pneumatic and pyrotechnic recovery loads, internal pressure, and local fastener loads.



3D-printed pattern



Glass-fibre female mould



CFRP tube prepreg layup

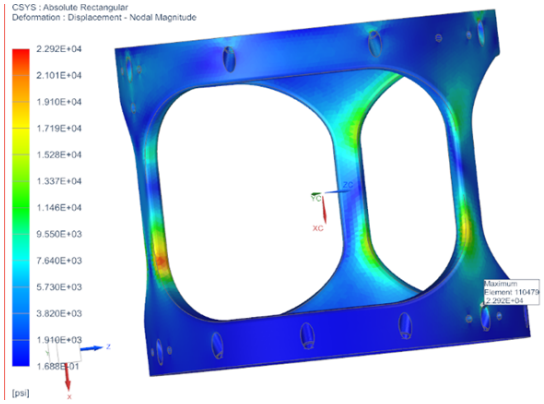
Figure 4: In-house tooling and layup: nose-cone pattern and mould, airframe tube on its mandrel.

Component	Governing load case	FoS	Verification
RADAX coupler	Vehicle bending	1.50	FEA
RADAX coupler	Pneumatic ejection	2.82	FEA
Payload plate	12 kN pneumatic ejection (122 MPa peak)	1.97	FEA
Engine retainer	Motor thrust (136.5 MPa peak)	1.77	FEA
Oxidizer tank	Internal pressure, MEOP 6.21 MPa	2.22	ASME sizing, FEA, hydrostatic proof test

Factors of safety on yield; Al 6061-T6 taken as 241 MPa (35 ksi) throughout this portfolio. Minimum required: 1.5.



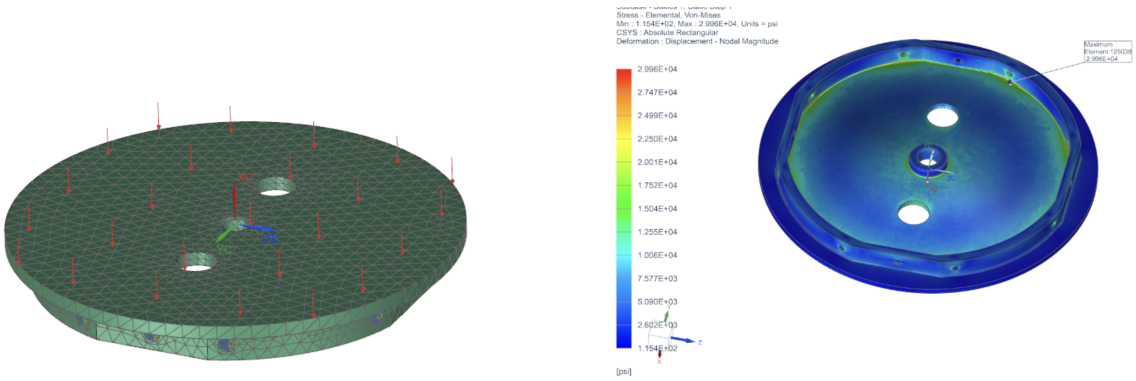
Machined RADAX coupler



Bending FEA (elemental stress)

Figure 5: RADAX coupler: governing bending case, FoS 1.50; pneumatic ejection case, FoS 2.82.

The payload plate shows the verification procedure used on the primary interfaces: stress was re-evaluated as element size was progressively reduced until the maximum converged at approximately 122 MPa, giving a factor of safety of 1.97 under the 12 kN ejection load.



Loads and boundary conditions *Converged stress field*

Figure 6: Payload plate under the 12 kN pneumatic ejection load: 122 MPa maximum, FoS 1.97.

Oxidizer tank integration

The oxidizer tank is part of the vehicle's primary structure, so its design had to satisfy pressure-vessel and airframe requirements together. It uses a machined Al 6061-T6 cylinder with Al 7075-T6 torispherical heads, sized to ASME pressure-vessel rules; FEA of the heads and pipe geometry located stress concentrations and identified where mass could be removed. The final tank has a 20.62 cm outer diameter, a 6.21 MPa maximum expected operating pressure (MEOP), 13 kg oxidizer capacity and an overall factor of safety of 2.22. It was CNC machined, sealed with radial fasteners and O-rings, hydrostatically proof-tested to 1.5×MEOP (≈9.3 MPa), and then passed four cold-flow and two hot-fire tests.



Figure 7: Manufactured oxidizer tank assembly and machined heads.

Vehicle-level stability and recovery

Structural mass, fin geometry and recovery architecture feed directly into flight behaviour, so they were iterated against vehicle-level models. In OpenRocket, the final configuration kept a minimum predicted stability margin of 1.5 calibers in winds up to about 17 mph while staying below the upper stability limit. For recovery, a dynamic model of the descent, including atmospheric density, drag-area growth during parachute inflation and deployment delay, was used to set a pre-apogee drogue trigger at approximately 30.6 m/s. Landing dispersion was then estimated by sweeping wind speed and direction.

Integration and flight

Parts from five processes (prepreg CFRP/GFRP, composite tooling, honeycomb sandwich, CNC machining and 3D printing) had to meet at shared interfaces. Repeated assembly rehearsals exposed integration problems before competition, and the final assembly was repeatedly completed in under

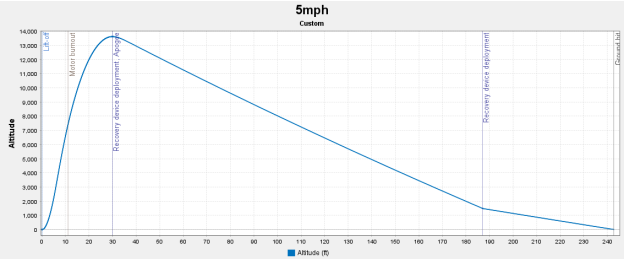


Figure 8: Vehicle-level flight model in OpenRocket.

one hour, confirming the purpose of the modular architecture. At Launch Canada the vehicle flew, placed 3rd in the Advanced category, and was recovered with no structural damage.



Figure 9: Integrated Project Luminor vehicle.

What this project shows. Ownership of a large mechanical system across its full life cycle, and a design method that does not rely on one tool: hand sizing, FEA with convergence checks, physical tests (hydrostatic proof, cold flow, hot fire) and finally flight each checked the others.

2 Classical Laminate Theory Design Tool

McGill Rocket Team • Individual project, developed as Aerostructures Subteam Lead

Composite structures require ply-level analysis because stiffness and strength depend on fibre orientation and stacking sequence. To size composite airframes and structural laminates, I developed an Excel-based classical laminate theory (CLT) calculator that converts material properties and flight loads into laminate stresses, failure indices and margins of safety.

The tool combines laminate stiffness calculations, six composite failure criteria, shell and panel buckling, and structural mass estimation. Its implementation was independently verified in Python and complemented by experimental testing of manufactured composite coupons.

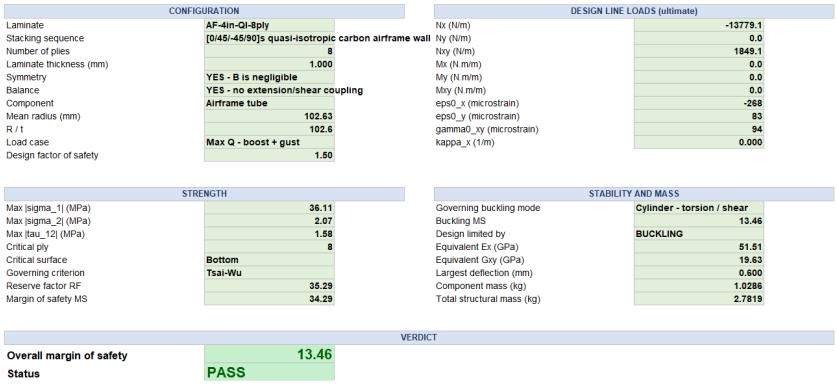
Calculator architecture

Thirteen linked worksheets form a traceable calculation pipeline:

$$\text{Materials} \rightarrow \text{Laminate } (ABD) \rightarrow \text{Loads}$$
$$\rightarrow \text{Ply stresses} \rightarrow \text{Failure and buckling} \rightarrow \text{Margins}$$

The material database includes unidirectional and woven carbon, glass and aramid composites. Users define ply orientations, thicknesses, geometry and applied loads (axial, bending, shear, torsion and pressure). The dashboard identifies the governing failure mode, minimum margin of safety, buckling capacity and structural mass.

Material inputs and stacking sequences can be modified without changing the underlying formulas, allowing rapid evaluation of alternative designs.



CONFIGURATION	
Laminate	AF-4in-QI-8ply
Stacking sequence	[0/45/-45/90]s quasi-isotropic carbon airframe wall
Number of plies	8
Laminate thickness (mm)	1.000
Symmetry	YES - B is negligible
Balance	YES - no extension/shear coupling
Component	Airframe tube
Mean radius (mm)	102.63
R / t	102.6
Load case	Max Q - boost + gust
Design factor of safety	1.50

DESIGN LINE LOADS (ultimate)	
Nx (N/m)	-13779.1
Ny (N/m)	0.0
Nxy (N/m)	1849.1
Mx (N.m/m)	0.0
My (N.m/m)	0.0
Mxy (N.m/m)	0.0
eps0_x (microstrain)	-268
eps0_y (microstrain)	83
gamma0_xy (microstrain)	94
kappa_x (1/m)	0.000

STRENGTH	
Max |sigma_1| (MPa)	36.11
Max |sigma_2| (MPa)	2.07
Max |tau_12| (MPa)	1.58
Critical ply	8
Critical surface	Bottom
Governing criterion	Tsai-Wu
Reserve factor RF	35.29
Margin of safety MS	34.29

STABILITY AND MASS	
Governing buckling mode	Cylinder - torsion / shear
Buckling MS	13.46
Design limited by	BUCKLING
Equivalent Ex (GPa)	51.51
Equivalent Gxy (GPa)	19.63
Largest deflection (mm)	0.600
Component mass (kg)	1.0286
Total structural mass (kg)	2.7819

VERDICT	
Overall margin of safety	13.46
Status	PASS

Figure 10: CLT dashboard summarizing applied loads, laminate configuration, governing failure mode and structural margins.

Theory and numerical verification

For an orthotropic ply, the reduced stiffness coefficients are

$$Q_{11} = \frac{E_1}{\Delta}, \quad Q_{22} = \frac{E_2}{\Delta}, \quad Q_{12} = \frac{\nu_{12} E_2}{\Delta}, \quad Q_{66} = G_{12}, \qquad \Delta = 1 - \nu_{12}\nu_{21}.$$

After transforming each ply stiffness matrix into laminate coordinates, the extensional, coupling and bending stiffnesses are obtained by integrating through the thickness:

$$A_{ij} = \sum_{k=1}^{n} \bar{Q}_{ij}^{(k)}(z_k - z_{k-1}),$$
$$B_{ij} = \frac{1}{2}\sum_{k=1}^{n} \bar{Q}_{ij}^{(k)}(z_k^2 - z_{k-1}^2),$$
$$D_{ij} = \frac{1}{3}\sum_{k=1}^{n} \bar{Q}_{ij}^{(k)}(z_k^3 - z_{k-1}^3).$$

Mid-plane strains and curvatures follow from

$$\begin{Bmatrix} N \\ M \end{Bmatrix} = \begin{bmatrix} A & B \\ B & D \end{bmatrix} \begin{Bmatrix} \varepsilon^0 \\ \kappa \end{Bmatrix}.$$

Ply stresses are recovered at the bottom, middle and top of each ply to capture through-thickness variations.

Independent verification. I developed a separate Python implementation using matrix transformations rather than the expanded stiffness equations used in Excel. Across 24 benchmark quantities, including the full ABD matrices, mid-plane strains, ply stresses and critical buckling loads, the two implementations agreed within 0.05%.

PLY DEFINITION (Input)					GEOMETRY (m)	
Ply #	Material	Angle (deg)	t override (mm)	t used (mm)	t (m)	z_bot (m)
1	CF-UD-T700-Epoxy	0.0		0.125	0.000125	-0.000500
2	CF-UD-T700-Epoxy	45.0		0.125	0.000125	-0.000375
3	CF-UD-T700-Epoxy	-45.0		0.125	0.000125	-0.000250
4	CF-UD-T700-Epoxy	90.0		0.125	0.000125	-0.000125
5	CF-UD-T700-Epoxy	90.0		0.125	0.000125	0.000000
6	CF-UD-T700-Epoxy	-45.0		0.125	0.000125	0.000125
7	CF-UD-T700-Epoxy	45.0		0.125	0.000125	0.000250
8	CF-UD-T700-Epoxy	0.0		0.125	0.000125	0.000375

ABD STIFFNESS MATRIX

	eps0_x	eps0_y	gamma0_xy	kappa_x	kappa_y	kappa_xy
Nx (N/m)	5.705E+07	1.778E+07	0.000E+00	-9.095E-13	2.842E-14	0.000E+00
Ny (N/m)	1.778E+07	5.705E+07	1.863E-09	2.842E-14	-6.821E-13	0.000E+00
Nxy (N/m)	0.000E+00	1.863E-09	1.963E+07	0.000E+00	0.000E+00	-2.842E-14
Mx (N.m/m)	-9.095E-13	2.842E-14	0.000E+00	7.974E+00	1.246E+00	4.971E-01
My (N.m/m)	2.842E-14	-6.821E-13	0.000E+00	1.246E+00	2.008E+00	4.971E-01
Mxy (N.m/m)	0.000E+00	0.000E+00	-2.842E-14	4.971E-01	4.971E-01	1.400E+00

Figure 11: Laminate builder showing ply orientations, thicknesses and calculated ABD stiffness matrix.

Failure and stability analysis

Six failure criteria are evaluated at every ply surface: maximum stress, maximum strain, Tsai–Hill, Tsai–Wu, Hashin and Puck. The calculator identifies the governing ply and failure mode and computes the corresponding reserve factor.

For Tsai–Wu, the reserve factor is determined from the positive root of the quadratic interaction equation:

$$a\lambda^2 + b\lambda - 1 = 0, \qquad \mathrm{RF} = \frac{-b + \sqrt{b^2 + 4a}}{2a},$$

$$a = F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\tau_{12}^2 + 2F_{12}\sigma_1\sigma_2,$$
$$b = F_1\sigma_1 + F_2\sigma_2.$$

This avoids the potentially non-conservative approximation RF = 1/FI when linear stress-interaction terms are present.

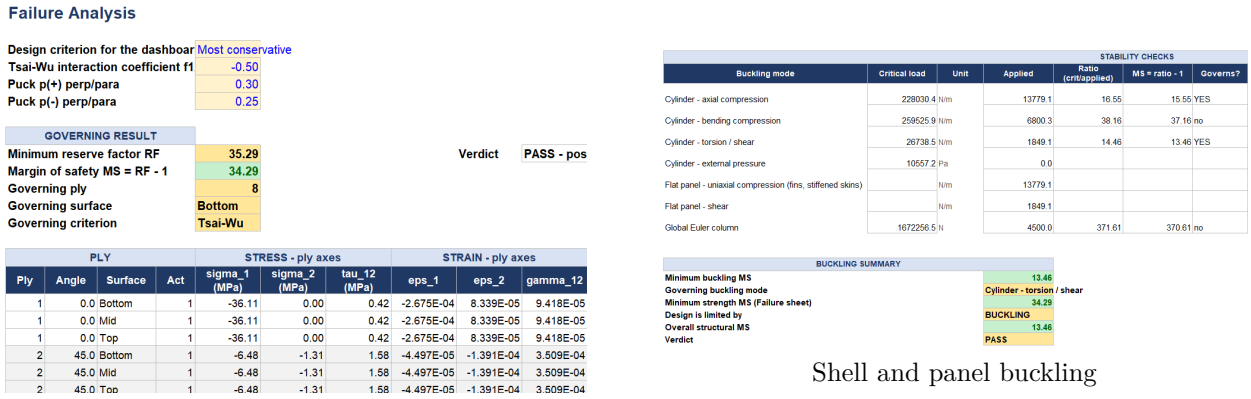
For cylindrical shells under axial compression, the classical orthotropic buckling load is

$$N_{x,\mathrm{cr}}^{\mathrm{cl}} = \frac{2}{R}\sqrt{D_{11}\left(A_{11} - \frac{A_{12}^2}{A_{22}}\right)}.$$

A NASA SP-8007 empirical knockdown factor accounts for geometric imperfections:

$$\phi = \frac{1}{16}\sqrt{\frac{R}{t}}, \qquad \gamma = 1 - 0.901(1 - e^{-\phi}).$$

By distinguishing strength-limited from buckling-limited designs, the tool guides laminate optimization: increasing reinforcement in the critical direction for strength, or bending stiffness for stability.



Failure Analysis

Design criterion for the dashboar	Most conservative
Tsai-Wu interaction coefficient f1	-0.50
Puck p(+) perp/para	0.30
Puck p(-) perp/para	0.25

GOVERNING RESULT	
Minimum reserve factor RF	35.29
Margin of safety MS = RF - 1	34.29
Governing ply	8
Governing surface	Bottom
Governing criterion	Tsai-Wu

Verdict: PASS - pos

PLY				STRESS - ply axes			STRAIN - ply axes		
Ply	Angle	Surface	Act	sigma_1 (MPa)	sigma_2 (MPa)	tau_12 (MPa)	eps_1	eps_2	gamma_12
1	0.0	Bottom	1	-36.11	0.00	0.42	-2.675E-04	8.339E-05	9.418E-05
1	0.0	Mid	1	-36.11	0.00	0.42	-2.675E-04	8.339E-05	9.418E-05
1	0.0	Top	1	-36.11	0.00	0.42	-2.675E-04	8.339E-05	9.418E-05
2	45.0	Bottom	1	-6.48	-1.31	1.58	-4.497E-05	-1.391E-04	3.509E-04
2	45.0	Mid	1	-6.48	-1.31	1.58	-4.497E-05	-1.391E-04	3.509E-04
2	45.0	Top	1	-6.48	-1.31	1.58	-4.497E-05	-1.391E-04	3.509E-04

Ply-level failure indices

STABILITY CHECKS

Buckling mode	Critical load	Unit	Applied	Ratio (crit/applied)	MS = ratio - 1	Governs?
Cylinder - axial compression	228030.4	N/m	13779.1	16.55	15.55	YES
Cylinder - bending compression	259525.9	N/m	6800.3	38.16	37.16	no
Cylinder - torsion / shear	26738.5	N/m	1849.1	14.46	13.46	YES
Cylinder - external pressure	10557.2	Pa	0.0			
Flat panel - uniaxial compression (fins, stiffened skins)		N/m	13779.1			
Flat panel - shear		N/m	1849.1			
Global Euler column	1672256.5	N	4500.0	371.61	370.61	no

BUCKLING SUMMARY	
Minimum buckling MS	13.46
Governing buckling mode	Cylinder - torsion / shear
Minimum strength MS (Failure sheet)	34.29
Design is limited by	BUCKLING
Overall structural MS	13.46
Verdict	PASS

Shell and panel buckling

Figure 12: Failure and stability calculations identifying the governing structural margins.

Design validation and flight performance

For the fins, aerodynamic bending was the governing loading concern. Their design incorporated conservative thicknesses to account for uncertainty associated with aged prepreg, while the CLT framework enabled evaluation of ply stresses and structural margins.

The project therefore brought together two complementary levels of verification and validation:

- **Numerical:** 24 independent Python benchmarks agreed within 0.05%, verifying the calculator's implementation.
- **Flight:** The selected composite structures completed the mission without observed structural damage.

What this project shows. An end-to-end composite structural design workflow combining classical laminate theory, automated failure and buckling analysis, independent numerical verification and flight application. The resulting tool provides a reusable foundation for sizing and optimizing future rocket structures.

3 Structural Redesign: Load-Path and Topology Optimization

McGill Rocket Team • Aerostructures Subteam Lead • Redesign for the team's next, higher-apogee vehicle

Two flight-critical parts were redesigned with two different strategies. For the recovery interface, the load path itself was changed so that material which carried no load could be removed. For the propulsion cage, the architecture was kept and topology optimization was used to redistribute material toward the dominant bending load, followed by a manual redesign for three-axis machining. Both designs had to keep their existing interfaces to the propulsion and recovery hardware and meet the team's minimum factor of safety of 1.5.

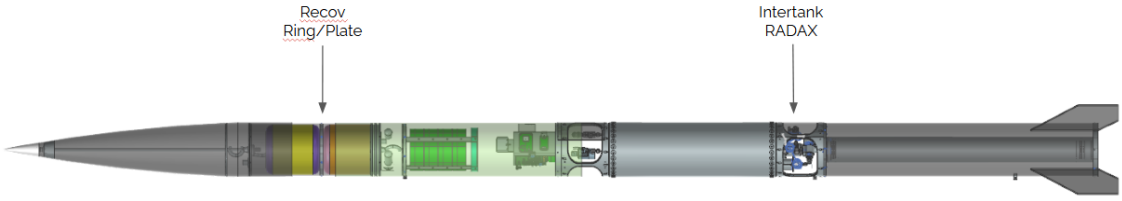
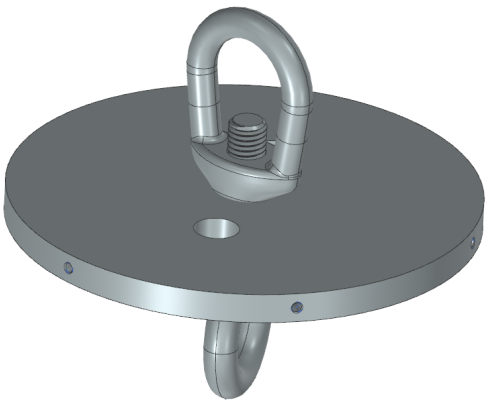


Figure 13: Location of the two redesigned structures within the vehicle.

3.1 Recovery interface: removing material that carried no load

The original interface was a central plate with two heavy eyebolts, one each for the main and drogue parachutes; the assembly weighed 3.4713 lbm (1.575 kg). The centre of the plate lay off the primary load path. The redesign carries the harness loads directly into a machined aluminium ring with two integral attachment regions and an open centre; two carbon-fibre plates bonded to opposite faces retain the parachute assembly so it cannot pass through the opening.

The ring weighs 0.4075 lbm and the two plates 0.1 lbm together, so the complete interface weighs 0.5075 lbm (0.230 kg): 85.4% lighter than the assembly it replaced.





Original: plate and two eyebolts, 1.575 kg Redesign: ring and CFRP plates, 0.230 kg

Figure 14: Original and redesigned recovery interfaces.

Verification

The ring was analysed under two recovery load cases with three progressively refined meshes and local mesh controls:

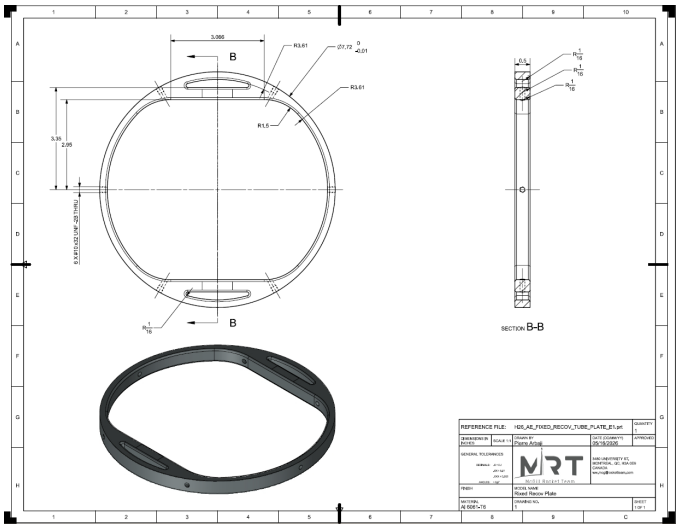


Figure 15: Manufacturing drawing used for CNC fabrication of the recovery ring.

Table 1: Recovery-ring mesh convergence.

Load (lbf)	Global mesh (in)	Local control (in)	Elemental stress (psi)	Nodal stress (psi)
1962	0.20	0.10	17690	25970
1962	0.12	0.06	17510	21780
1962	0.08	0.03	18610	21290
2700	0.20	0.10	3631	4663
2700	0.12	0.06	3742	4417
2700	0.08	0.03	3882	4346

For the governing 1962 lbf (8.73 kN) case, nodal stress changed by 2.3% between the two finest meshes and elemental stress by 6.3%. At the finest mesh the two still differ by about 14%, indicating a local gradient at the peak, so the margin is based on the higher nodal value:

$$FoS_y = \frac{35\,000}{21\,290} = 1.64 \;\geq\; 1.5.$$

3.2 Intertank RADAX cage: topology optimization under a manufacturing constraint

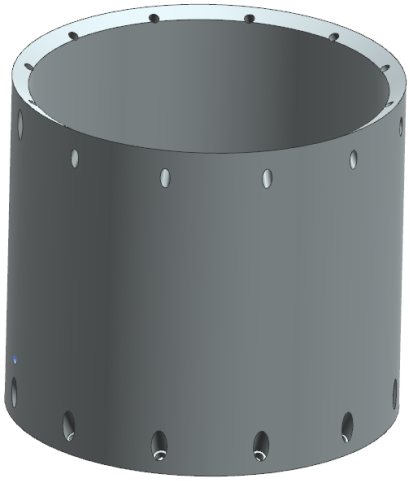
The cage sits between the thrust chamber and the oxidizer tank. It carries the primary thrust and bending loads while leaving access to the propulsion valves. The original Al 6061-T6 thin-lattice cage met the current requirements, but its bending factor of safety was 1.50, exactly the team minimum. The next vehicle flies to a higher apogee, which raises the bending load, so bending became the design driver, with mass and manufacturability as secondary objectives. The other three cases already had factors of safety of more than 4.

Table 2: Load cases and margins of the original cage.

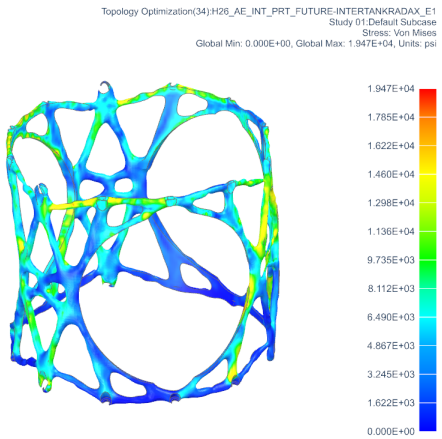
Load case	Load (N)	FoS	Governing condition
Thrust	6204	5.61	Launch / maximum acceleration
Bending	2211	1.50	Maximum normal aerodynamic load
Buckling	6204	35.4	Launch / maximum acceleration
Tension	8558	4.00	Worst-case parachute shock

Topology optimization as a load-path guide

The interfaces and load-introduction regions were frozen as non-design space. On the remaining volume, the optimization minimized mass subject to a factor of safety of 2.0 under the bending case, with a mirror-symmetry constraint and a cyclic-symmetry constraint forcing three identical sectors. The result was used as a guide to the load paths, not as a geometry: material was kept around the interfaces and along the dominant paths, and the part was then redrawn by hand to meet manufacturing and integration constraints.



Design and non-design space



Topology optimization result

Figure 16: Design space and topology optimization of the structural cage: minimum mass, FoS 2.0 in bending, three identical sectors.

Design for manufacturing

The original lattice required five-axis machining. The redesigned openings were shaped so the principal features can be cut on a conventional three-axis CNC setup while following the load paths from the optimization. Opening sizes need to be set with the propulsion team: large enough for valve access, no larger.

Verification

Under the governing bending case, the redesigned cage reached a maximum stress of

$$\sigma_{\max} = 1.663 \times 10^4 \text{ psi (114.7 MPa)}, \qquad FoS_y = \frac{35\,000}{16\,630} = 2.10,$$

above the 2.0 target. Thrust, buckling and tension were not re-analysed for the new geometry; the redesign addressed bending only.

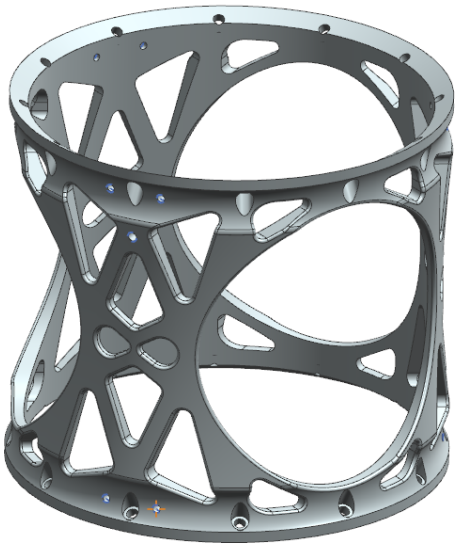
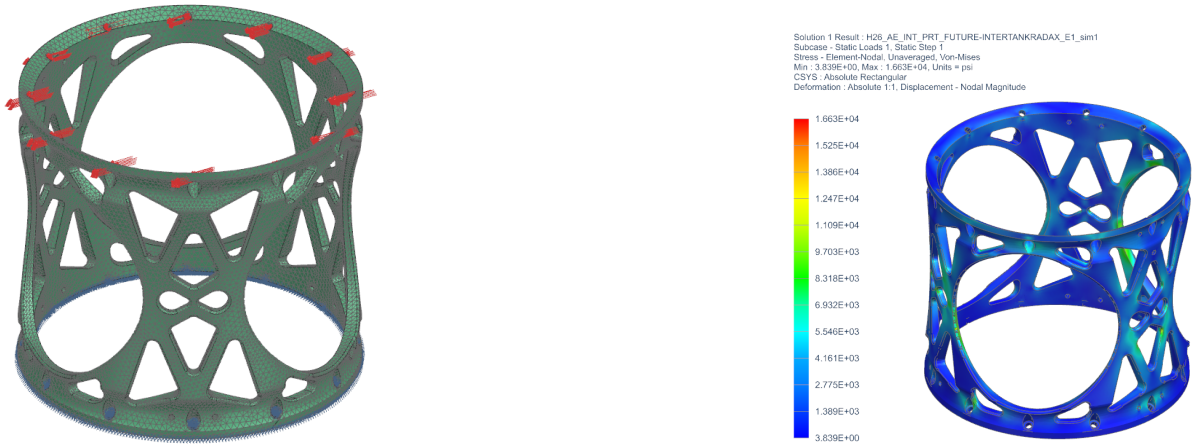


Figure 17: Final cage geometry.



Loads and constraints Stress result, redesigned cage

Figure 18: Bending FEA of the redesigned cage: 114.7 MPa maximum, FoS 2.10.

Results

Table 3: Summary of the two redesigns.

Component	Original mass	New mass	Mass change	Factor of safety
Recovery interface	1.575 kg	0.230 kg	−85.4%	1.64
Propulsion cage	0.854 kg	0.703 kg	−17.7%	Bending 1.50 → 2.10

What this project shows. Two routes to a lighter structure: change the load path when material is doing no work, and redistribute material when the architecture is sound but the margin is not. In both cases the optimizer and the FEA informed the design; manufacturing and integration constraints set its final form.

4 Research: Electro-Thermal Supercoiled Polymer Muscles for Bistable Metamaterials

Architected Metamaterials Group, McGill • Undergraduate Research Assistant, Mar 2026 – present

Research context. Reprogrammable bistable metamaterials switch between two mechanically stable configurations through snap-through instability. Controlled reconfiguration requires an actuator capable of generating sufficient force and displacement while fitting within the structure itself.

The target was a laser-cut nylon-foam bistable unit requiring a 2.0 N actuation force and 15 mm displacement to trigger switching. I developed a dual-fibre supercoiled polymer (SCP) artificial muscle driven by Nichrome resistance heating, characterized its electro-thermal performance and failure limits, and designed and implemented its mechanical integration into the physical metamaterial.

Actuator design and fabrication

A dual-fibre plied configuration was selected to increase load-bearing capacity while maintaining a compact geometry. Actuation relies on Joule heating: a helically wrapped Nichrome wire heats the polymer, producing axial contraction.

The fabrication procedure consisted of four stages:

1. **Coiling:** Two nylon monofilament strands were individually twisted under a 290 g (2.84 N) hanging load until full-length self-coiling.
2. **Plying:** The coiled strands were twisted together around a common axis under a 500 g (4.90 N) load to form the dual-fibre actuator.
3. **Thermal annealing:** The muscle was stretched to 110% of its relaxed length and annealed at 160°C for 1 h. After cooling, it was extended a further 33% and annealed again for 1 h at 160°C to stabilize its coiled geometry.
4. **Resistive heater:** Bare 34 AWG Nichrome 80 wire was wound around the muscle at an approximately 1.2 mm pitch, providing distributed heating while avoiding contact between adjacent turns.

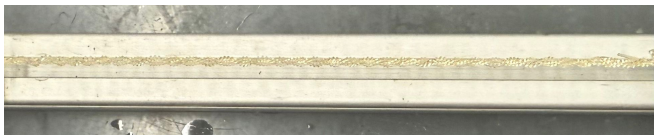


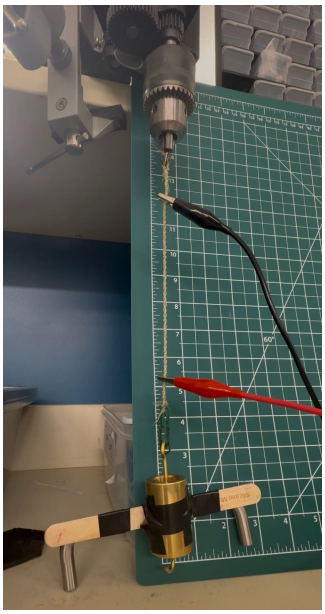
Figure 19: Fabricated dual-fibre SCP muscle prior to Nichrome heater integration.

Electro-thermal characterization

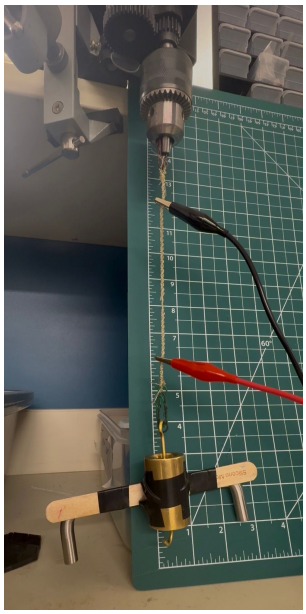
The actuator was suspended vertically under a constant 290 g (2.84 N) dead load, with an active heated length of 158.75 mm. A stepped voltage sweep from 4.0 to 8.5 V was used to characterize contraction and identify the operating limits.

The muscle achieved a maximum contraction of 19.05 mm (0.75 in) before thermal failure at 8.5 V. The resulting actuation strain, output force and mechanical work were

$$\varepsilon_a = \frac{\Delta L}{L_0} = \frac{19.05}{158.75} = 12.0\%,$$
$$F_L = mg = 0.290(9.81) = 2.84 \text{ N},$$
$$W_{\text{out}} = F_L \Delta L = 2.84(0.01905) = 0.054 \text{ J}.$$



Before actuation



After contraction

Figure 20: Electro-thermal contraction of the SCP muscle under a constant 2.84 N load, achieving a 19.05 mm stroke.

Performance metric	**Requirement**	**Measured**
Actuation force	2.00 N	2.84 N
Contraction stroke	15.0 mm	19.05 mm
Actuation strain	—	12.0%
Mechanical work	—	0.054 J
Rupture voltage	—	8.5 V
Conservative drive limit	—	≤ 7.5 V

Table 4: Measured actuator performance relative to bistable metamaterial requirements.

The actuator exceeded the required force by 42% and the required stroke by 27%, demonstrating that the developed muscle could satisfy the mechanical prerequisites for bistable switching.

Failure analysis and operating limits

At 8.5 V, the polymer core ruptured near the centre of the heated region, where local temperatures exceeded approximately 220°C. Under the sustained 2.84 N tensile load, localized thermal softening and loss of mechanical strength were identified as the principal failure mechanisms.

For approximately constant electrical resistance, heating power increases according to

$$P = \frac{V^2}{R}.$$

The voltage sweep therefore produced increasingly severe thermal loading at higher drive levels. Based on the observed behaviour, a conservative continuous operating limit of 7.5 V was established for the tested actuator configuration. This failure analysis identified the need to control heating power and temperature rather than voltage alone, particularly for repeated actuation cycles.

Mechanical integration and physical prototype

Following actuator characterization, the SCP muscle was physically integrated into the hollow interior of the bistable metamaterial unit cell. The muscle ends pass through the end plates and are secured by epoxying the muscle to the plates which had an angled hole to help the epoxy sit nicely. The anchoring system resisted axial loads exceeding 10 N, providing a capacity greater than 3.5 times the demonstrated actuation load.



Figure 21: Physical integration of the SCP muscle within the bistable metamaterial prototype.

Research outcomes and next steps

The work established a fabrication and integration methodology for electrically driven SCP actuators in bistable metamaterials. The developed muscle achieved 12% contraction against a 2.84 N load, exceeding both the force and stroke requirements of the target mechanism. Tensile testing of the anchoring system confirmed its capacity to transmit the demonstrated actuator load, and physical integration into the metamaterial was completed.

Current research focuses on closed-loop pulse-width-modulated (PWM) drive control to limit overheating, repeated actuation and fatigue characterization, and thermal management to reduce cooling and reset times. Full-assembly switching tests will assess the coupled response of the actuator and bistable structure.

What this project shows. Experimental research spanning actuator design, polymer processing, electro-thermal characterization, failure analysis and physical system integration. The project demonstrates the development of a compact artificial muscle from material fabrication through mechanical testing and assembly into a functional metamaterial prototype.

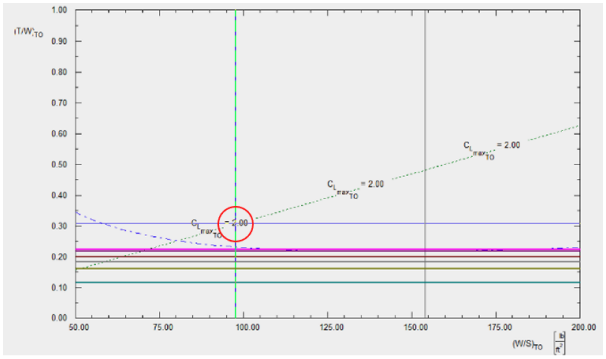
5 Aircraft Conceptual Design: Long-Range Business Jet

McGill AERO 520 (aircraft design) • Team of four • My responsibilities: Led Structural Layout Design

Conceptual and preliminary design of an ultra-long-range business jet for entry into service around 2035, answering a request for proposal (RFP) aimed at the market segment of the Gulfstream G700 and Bombardier Global 7500. Key requirements: 8,000 nmi range with 10% fuel reserves; Mach 0.85 cruise minimum (0.92 target); takeoff and landing field lengths under 6,000 ft (sea level, ISA); FAA Part 25 with flight into known icing; eight passengers, bedroom, shower, galley and three crew.



Final configuration



Matching plot and design point

Figure 22: Configuration and performance-constraint sizing.

Sizing. Takeoff weight was estimated with Roskam's regression method using justified 2035 assumptions ($c_j = 0.40$, $L/D = 14$). In the constraint (matching) plot, stall speed and takeoff distance governed; the design point was placed at their intersection to minimize wing area: $W/S = 97.57$ lb/ft^2, $T/W = 0.31$, $S_w = 1{,}310.25$ ft^2.

Configuration. XFOIL/AAA iterations showed an SC(2)-0410 wing could not reach the required cruise $C_L \approx 0.44$; a more cambered SC(2)-0610 root gave $C_L \approx 0.45$. The wing has 35° quarter-chord sweep (critical Mach), taper 0.30 and −5° washout (root-first stall), with a T-tail sized by volume coefficients. Class II weights from three methods (discarding the lowest estimate for conservatism) converged to an MTOW of 95,260 lb and a takeoff thrust requirement of 29,676 lbf, met by two GE Passport engines rubber-scaled by 0.784.

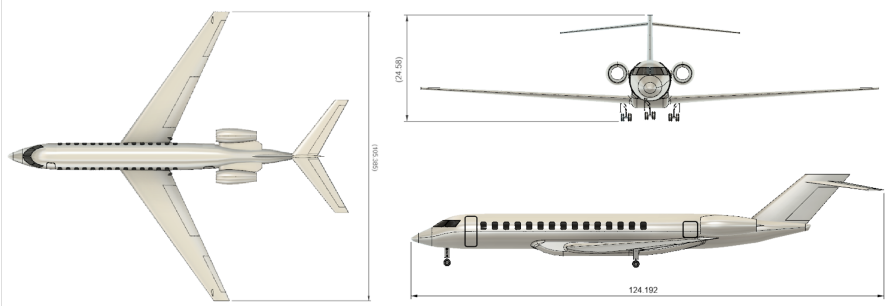
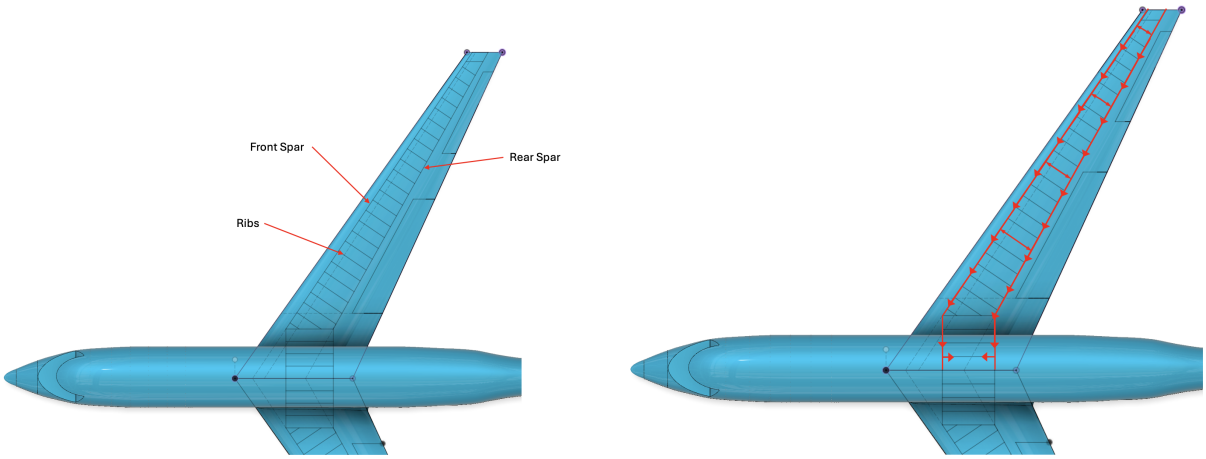


Figure 23: Three-view: 124.2 ft long, 105.53 ft span, 24.58 ft tall.

Structural layout. Fuselage frames at 18 in and longerons at 9 in (tightened to 12 in frames

around the windshield); flat forward and domed aft pressure bulkheads, with the aft bulkhead behind the baggage bay so baggage stays pressurized. The wing box uses spars at 15% and 60% chord, the rear spar placed forward of Roskam's recommended range to make room for flaps and ailerons, with ribs every 24 in, bolted to reinforced fuselage frames.



Wing spars and ribs Load path: skin → ribs → spars/stringers → frames

Figure 24: Wing structure and load path.

Performance and stability. Takeoff and landing were integrated segment by segment in MATLAB with velocity-dependent lift, drag and thrust; the balanced field length is the intersection of the accelerate-stop and accelerate-go curves. An initial elevator-trim design could not trim the full CG range (26.9–38.2% MAC) at high lift coefficients; switching to a trimmable horizontal stabilizer solved it in cruise and landing.

Table 5: Performance against the RFP.

Requirement	Target	Achieved	Status
Takeoff field length	$\leq$ 6,000 ft	5,234 ft	Met
Landing field length	$\leq$ 6,000 ft	5,206 ft	Met
Cruise Mach (minimum)	0.85	0.85	Met
Cruise Mach (target)	0.92	$M_{crit} = 0.876$	Not met
Design range	8,000 nmi	7,849 nmi	1.9% short
Cabin accommodations	RFP list	All included	Met

Engineering takeaway. The shortfalls were traced rather than hidden: range is limited by cruise L/D (13.28 from AAA; 14 would give 8,275 nmi) and the Mach 0.92 target by the wing and tail critical Mach numbers (0.876 and 0.883), pointing to airfoil tailoring as the next iteration. A sensitivity study showed the landing distance was driven mainly by the assumed braking friction coefficient.